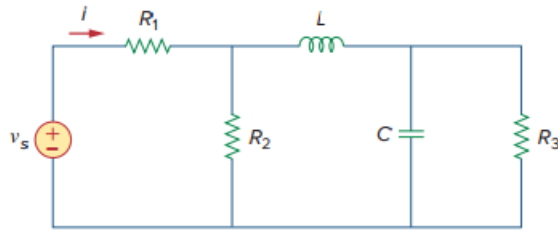


Problem

Using Fig. 9.43, design a problem to help other students better understand impedance.

Figure 9.43:



Step-by-step solution

Step 1 of 5

Given circuit can be represented as

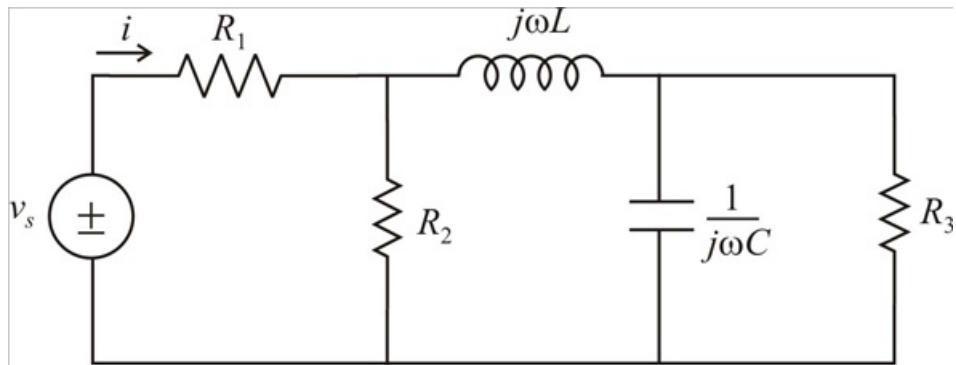


Figure (1)

Step 2 of 5

Above circuit can be represented as ,

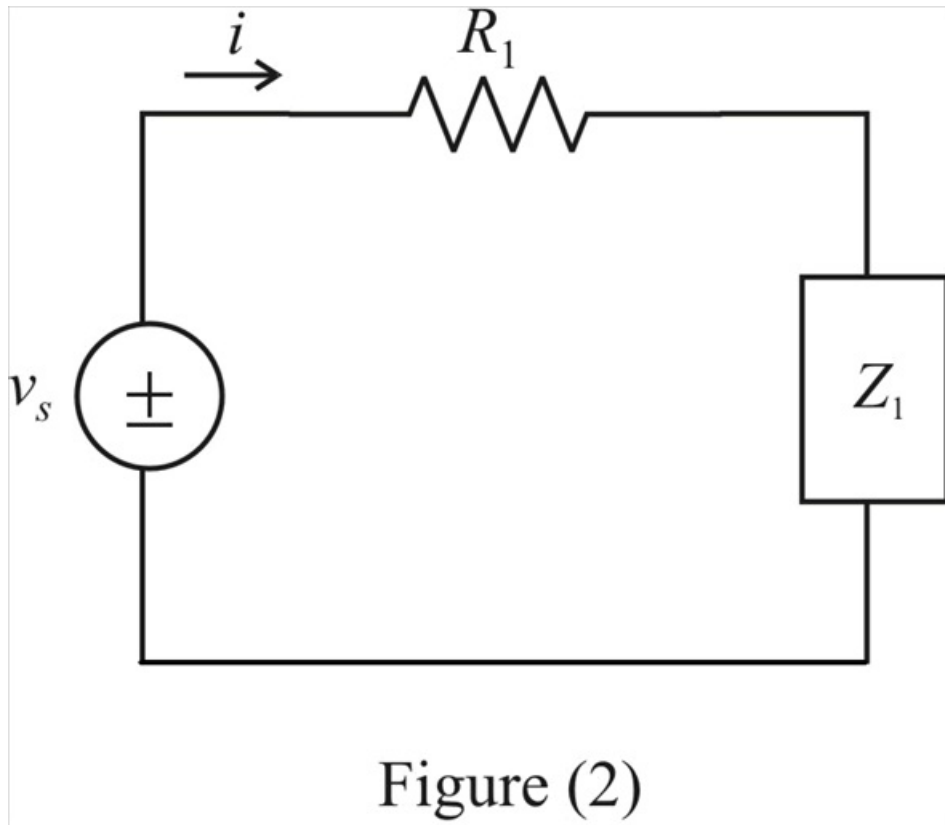


Figure (2)

Step 3 of 5

Where $Z_1 = R_2 \parallel \left[j\omega L + \left(\frac{1}{j\omega C} \parallel R_3 \right) \right]$

$$Z_1 = R_2 \parallel \left[j\omega L + \frac{\frac{R_3}{j\omega C}}{R_3 + \frac{1}{j\omega C}} \right]$$

$$Z_1 = R_2 \parallel \left[j\omega L + \left(\frac{R_3}{1 + j\omega C R_3} \right) \right]$$

$$Z_1 = R_2 \parallel \left[\frac{(j\omega L + j^2 \omega^2 R_3 LC) + R_3}{1 + j\omega C R_3} \right]$$

Step 4 of 5

$$Z_1 = R_2 \parallel \left[\frac{j\omega L + R_3 - \omega^2 R_3 LC}{1 + j\omega C R_3} \right]$$

$$Z_1 = \frac{R_2 \times \left[\frac{j\omega L + R_3 - \omega^2 R_3 LC}{1 + j\omega C R_3} \right]}{R_2 + \left[\frac{j\omega L + R_3 - \omega^2 R_3 LC}{1 + j\omega C R_3} \right]}$$

$$Z_1 = \frac{R_2 \times [j\omega L + R_3 - \omega^2 R_3 LC]}{R_2 + j\omega C R_2 R_3 + j\omega L + R_3 - \omega^2 R_3 LC}$$

Step 5 of 5

From figure (2)

$$i = \frac{v_s}{R_1 + Z_1}$$

Substitute Z_1 in the above equation

$$i = \frac{v_s}{R_1 + \left[\frac{R_2 (j\omega L + R_3 - \omega^2 R_3 LC)}{R_2 + j\omega C R_2 R_3 + j\omega L + R_3 - \omega^2 R_3 LC} \right]}$$

$$i = \frac{v_s (R_2 + R_3 + j\omega C R_2 R_3 + j\omega L - \omega^2 R_3 LC)}{R_1 R_2 + j\omega C R_1 R_2 R_3 + j\omega R_1 L + R_1 R_3 - \omega^2 R_1 R_3 LC + j\omega R_2 L + R_2 R_3 - \omega^2 R_2 R_3 LC}$$

$$\therefore i = \boxed{\frac{v_s (R_2 + R_3 + j\omega C R_2 R_3 + j\omega L - \omega^2 R_3 LC)}{R_1 R_2 + R_2 R_3 + R_1 R_3 + j\omega C R_1 R_2 R_3 + j\omega L (R_1 + R_2) - \omega^2 R_3 LC (R_1 + R_2)}}$$